

Regional dynamic traffic assignment framework for multi-reservoir MFD models

Sérgio Batista^{*1} and Ludovic Leclercq^{†1}

¹Univ. Lyon, IFSTTAR, ENTPE, LICIT, F-69675, Lyon, France

March 15, 2018

1 Extended Abstract

2 The first ideas of an aggregated traffic modeling were introduced by Godfrey (1969)
3 and later revisited by Daganzo (2007) and Geroliminis & Daganzo (2008). For
4 this type of modeling, the city is divided into reservoirs (Fig. 1) where the traffic
5 conditions are approximately homogeneous and are given by a well-defined low
6 scatter Macroscopic Fundamental Diagram (MFD). The MFD is a relationship
7 between the aggregated flow q and aggregated density k . The traffic dynamic
8 inside a reservoir is governed by the evolution of the vehicles accumulation n at a
9 given time instant t Daganzo (2007):

$$\frac{dn}{dt} = Q_{in} - Q_{out}, t > 0 \quad (1)$$

10 where Q_{in} and Q_{out} are the inflow and outflow functions, respectively.

11 Depending on the assumptions made on Q_{out} , two MFD-based models can
12 be distinguished in the literature: the accumulation-based (Daganzo, 2007); and
13 the trip-based (Arnott, 2013; Lamotte & Geroliminis, 2016; Mariotte et al., 2017).
14 In this work, we focus our attention on the trip-based model, where the MFD
15 dynamics is entirely described by the vehicle trip length L :

$$L = \int_{t-T(t)}^t V(n(s))ds \quad (2)$$

16 where $T(t)$ is the travel time of the vehicle exiting at t . Our numerical implemen-
17 tation of this model is based on Mariotte et al. (2017).

18 One key ingredient for a proper trip-based MFD simulation is a decent es-
19 timation of trip lengths. In fact, different vehicles might travel different distances
20 inside the same reservoir. In Fig. 1 (a), we show an example of a microscopic net-
21 work where there are represented three microscopic trips. A microscopic trip is

*✉ sergiofilipe.assuncaobatista@entpe.fr

†✉ ludovic.leclercq@entpe.fr

1 defined by a sequence of links. Two microscopic trips are correlated if they share
 2 links in common. This is represented by the red links shown in Fig. 1 (a). But,
 3 when we scale from microscopic to the regional network, we no longer have access
 4 to the topology information of the real network (Fig. 1 (b)). We focus instead on
 5 the set of reservoirs that define the regional network as shown in Fig. 1 (b). In the
 6 regional network, microscopic origin-destination (od) pairs correspond to macro-
 7 scopic Origin-Destination (OD) reservoirs; and each microscopic trip corresponds
 8 to a regional path. A regional path is defined by the sequence of crossed reservoirs
 9 from O to D. In Fig. 1 (a), the green microscopic trips define the same regional
 10 path. But, they have different trip lengths inside each reservoir that is crossed. In
 11 Fig. 1 (b), the green and blue regional paths are correlated since they cross two
 12 reservoirs in common. In Fig. 1 (c), we show a zoom of the gray reservoir with a
 13 well-defined MFD function. We denote L_1 and L_2 as the trip lengths of the green
 14 and blue regional paths inside the gray reservoir, respectively. There are two main
 15 differences between the correlation on the microscopic network and the one on the
 16 regional network. First, the correlation between the green and blue regional paths
 17 inside the gray reservoir are captured by the homogeneous speed assumption of
 18 the MFD-based models. That is, a vehicle that enters the gray reservoir and is
 19 traveling on the blue regional path, will automatically affect the travel times of all
 20 vehicles inside this reservoir, independent of the regional path they are traveling.
 21 Second, a regional path is characterized by a distribution of trip lengths inside each
 22 reservoir that it crosses. This is because one regional path can be defined by several
 23 microscopic trips with distinct trip lengths.

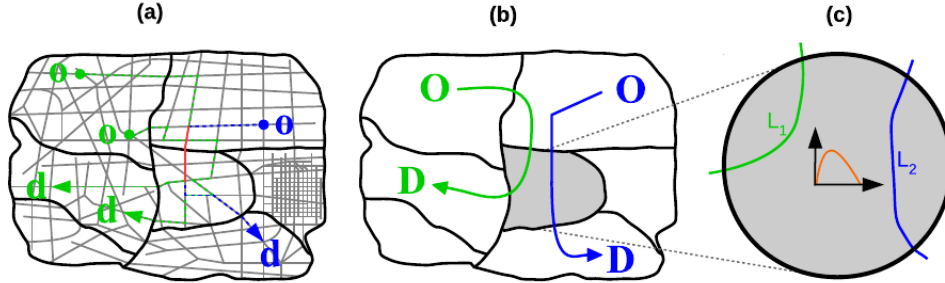


Fig. 1 – (a) Microscopic network with three microscopic routes. Two of these
 microscopic routes are correlated as shown by the red links. The macroscopic
 network is also represented. (b) Macroscopic network with the corresponding
 macro-paths to the green and blue microscopic routes. (c) Gray reservoir
 with a well-defined MFD function, that is crossed by the green and blue
 macro-paths, each with a corresponding trip length L_1 and L_2 .

24 Few attention has been paid to the dynamic traffic assignment on the MFD-
 25 based models context, i.e. the characterization of regional paths and the calcula-
 26 tion of the path flow distributions. Up to the authors best knowledge, the most
 27 advanced dynamic traffic assignment framework for the MFD-based models is dis-
 28 cussed by Yildirimoglu & Geroliminis (2014) (for the Stochastic User Equilibrium)
 29 and Yildirimoglu et al. (2015) (for the System Optimum). Yildirimoglu & Gerolim-
 30 inis (2014) discuss a regional dynamic traffic assignment framework that considers

trip lengths that are explicitly calculated in an iterative way. The stochastic network loading is based on the Multinomial Logit formulation.

In this work, we discuss a methodology for a regional-based dynamic traffic assignment that accounts for explicitly calculated trip lengths, to solve the Deterministic and Stochastic User Equilibrium (DUE and SUE). For this, we consider the utility function defined as:

$$U_p^{OD} = \sum_{r \in Y} \left(\frac{L_{rp}}{V_r(n_r)} \right) \delta_{rp} \quad (3)$$

$$\approx \sum_{r \in Y} \left(\frac{\overline{L_{rp}}}{\overline{V_r(n_r)}} + \epsilon_r(L_{rp}, V_r(n_r)) \right) \delta_{rp}, \forall p \in \Omega^{OD} \wedge \forall (O, D) \in W$$

where L_{rp} is the set of trip lengths of regional path p inside reservoir r ; $V_r(n_r)$ is the speed-MFD function of reservoir r ; δ_{rp} is a dummy variable that equals 1 if regional path p crosses reservoir r ; Y is the set of reservoirs that define the regional network; Ω^{OD} is the macroscopic choice set for the macroscopic origin-destination (OD) pair; and W is the set of all macroscopic OD pairs of the regional network. In Eq. 3 the term $\frac{\overline{L_{rp}}}{\overline{V_r(n_r)}}$ defines the deterministic part of the utility function. While, the term $\epsilon_r(L_{rp}, V_r(n_r))$ is the error term that depends on:

- L_{rp} that defines the distribution of trip lengths of regional path p inside reservoir r . Batista et al. (2018) investigates four approaches to properly define L_{rp} based on a set Γ of microscopic trips. The authors propose to filter these microscopic trips based on different levels of information of the regional network. We follow one of the approaches discussed in Batista et al. (2018), that considers a level of filtering at the regional path level. That is, all microscopic routes that travel inside a given reservoir and that define the same regional path p are considered to define L_{rp} .
- $v_r(n(t))$ is the speed-MFD function and yields the users the different perception of the traffic states inside reservoir r .

Note that, for the DUE, $\epsilon_r(L_{rp}, V_r(n_r))$ is set to 0.

To calculate the network equilibrium, accounting for these two uncertainty terms, we consider Monte Carlo simulations (Sheffi 1985). Both the DUE and SUE are formulated as fixed-point problems and are solved based on the Method of Successive Averages. We choose a $\frac{1}{k}$ descent step to ensure the good algorithm convergence. The idea of the Monte Carlo simulations is to sample trip lengths directly from L_{rp} and mean speeds from $v_r(n(t))$. We perform a large number (M) of samples. For each one, we solve deterministic problems where users minimize their perceived utility. The final choices, for each MSA descent loop, correspond to the averaging of all these local deterministic choices. As convergence criterion, we consider the Gap function and the number of violations (Sbayti et al., 2007). A maximum number of descent steps N_{max} is also fixed.

We discuss some preliminary results based on one implementation on a 1 reservoir network with entry queuing. The network is composed by 1 OD and two regional paths (Fig. 2). For the MFD-trip based simulation, we consider a parabolic MFD function with: critical jam $n_{jam} = 1000$ veh; critical production

$P_{critical} = 3000$ veh.m/s; and free-flow speed $u = 15$ m/s. The demand level is:
 0.3 veh/s between 0 and 1000 s; 1.0 veh/s between 1000 and 6000 s; and 0.3 veh/s
 between 6000 and 10000 s. For the convergence, we fix 0 for the maximum number of
 violations, a Gap tolerance of 0.01 and $N_{max} = 100$. For this test, we fix $L_2 = 1500$
 m and vary $L_1 \in [1300, 1700]$ as shown in Table 1. For the SUE calculations, the trip
 lengths are sampled for the two regional paths following a normal distribution. For
 the sampling of trip lengths for regional path 2, we consider a normal distribution
 with mean 1500 m and standard deviation σ_L . While, for the case of regional
 path 1, we consider a normal distribution with mean value as listed in Table 1 and
 standard deviation σ_L . We consider three different values for $\sigma_L = 50, 100, 200$ m.
 The calculated regional path flows for the DUE and three SUE cases are listed in
 Table 1.

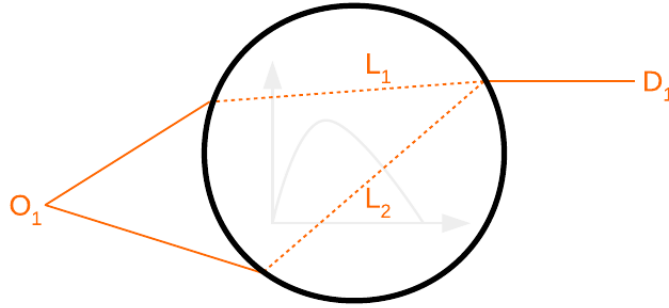


Fig. 2 – One reservoir test network with one OD and two regional paths.

We first analyze the DUE results, where $\epsilon_r(L_{rp}, V_r(n_r))$ is set to 0 (see Eq. 3).
 For $L_1 < L_2$, all users choose the minimum utility that corresponds to regional path
 1, since the trip length is smaller. Note that the MFD dynamics is the same for
 both regional paths and the only term that affects the utility function are the trip
 lengths. For $L_1 = L_2 = 1500$ m, users equally choose regional paths 1 and 2. For
 $L_1 > L_2$, all users choose regional path 2, as expected.

We now analyze the SUE case, considering first $\sigma_L = 50$. Initially, for
 $L_1 = 1300$ m, all users choose the regional path 1. Note that, the distribution of
 trip lengths for regional path 1 follows a $N(1300, 50)$ and for regional path 2 follows
 a $N(1500, 50)$. Since the standard deviation is small, the values that are sampled
 for the trip length distribution of regional path 1 are always inferior to the ones for
 regional path 2. Thus, all users choose regional path 1. But, as L_1 increases, users
 start to also choose regional path 2. For $L_1 = L_2 = 1500$ m, users also equally
 choose regional paths 1 and 2, as expected. For $L_1 > 1500$ m, the fraction of users
 that choose regional path 2 increases. This happens until $L_1 = 1650$, when all users
 choose regional path 1. A similar trend for switching from regional path 1 to 2 as
 L_1 increases is also observed.

In the extended version of this work, we will consider this one reservoir MFD
 model with entry queuing to: (i) analyze the algorithm convergence for both DUE
 and SUE; (ii) compare the SUE results against the Multinomial Logit model; (iii)
 analyze the DUE and SUE results considering 2 OD pairs. We will also analyze
 the implementation of this framework for both DUE and SUE calculations on a
 real network and considering a re-assignment procedure per periods during the

L_1	DUE		SUE ($\sigma_L = 50$)		SUE ($\sigma_L = 100$)		SUE ($\sigma_L = 200$)	
	Q_1	Q_2	Q_1	Q_2	Q_1	Q_2	Q_1	Q_2
1300	1.00	0.00	1.00	0.00	1.00	0.00	0.92	0.08
1350	1.00	0.00	1.00	0.00	0.98	0.02	0.85	0.15
1400	1.00	0.00	0.98	0.02	0.91	0.09	0.76	0.24
1425	1.00	0.00	0.94	0.06	0.84	0.16	0.69	0.31
1450	1.00	0.00	0.87	0.13	0.74	0.26	0.64	0.36
1475	1.00	0.00	0.73	0.27	0.64	0.36	0.57	0.43
1500	0.50	0.50	0.51	0.49	0.50	0.50	0.50	0.50
1525	0.00	1.00	0.28	0.72	0.38	0.62	0.43	0.57
1550	0.00	1.00	0.14	0.86	0.24	0.76	0.36	0.64
1575	0.00	1.00	0.07	0.93	0.17	0.83	0.30	0.70
1600	0.00	1.00	0.03	0.97	0.10	0.90	0.25	0.75
1650	0.00	1.00	0.00	1.00	0.03	0.97	0.15	0.85
1700	0.00	1.00	0.00	1.00	0.00	1.00	0.08	0.92

Tab. 1 – *Regional path flows for the DUE and SUE results and for the one reservoir MFD model with entry queuing. Q_1 and Q_2 are the flows of regional paths 1 and 2, respectively. Three values of σ_L are considered. The trip length $L_2 = 1500$ m is kept constant, while L_1 is increased from 1300 to 1700 m.*

1 simulation. For this test, we will simulate a morning peak.

2 Acknowledgments

3 This project is supported by the European Research Council (ERC) under the European
4 Union's Horizon 2020 research and innovation program (grant agreement No 646592 -
5 MAGnUM project). S. F. A. Batista also acknowledges funding support by the region
6 Auvergne-Rhône-Alpes (ARC7 Research Program).

7 References

- 8 Arnott, R. (2013), *A bathtub model of downtown traffic congestion*. Journal of Urban
9 Economics, 76, 110–121, [doi:10.1016/j.jue.2013.01.001](https://doi.org/10.1016/j.jue.2013.01.001)
- 10 Batista, S. F. A., Leclercq, L. & Geroliminis, N. (2018), *Trip length estimation for the ag-*
11 *gregated network models: scaling microscopic trips into reservoirs*. submitted to Trans-
12 *portation Research Part B: Methodological*.
- 13 Daganzo, C. (2007), *Urban gridlock: Macroscopic modeling and mitigation*
14 *approaches*. Transportation Research Part B: Methodological, 41, 49–62,
15 [doi:10.1016/j.trb.2006.03.001](https://doi.org/10.1016/j.trb.2006.03.001)
- 16 Geroliminis, N. & Daganzo, C. (2008), *Existence of urban-scale macroscopic fundamental*
17 *diagrams: Some experimental findings*. Transportation Research Part B: Methodologi-
18 *cal*, 42, 759–770, [doi:10.1016/j.trb.2008.02.002](https://doi.org/10.1016/j.trb.2008.02.002)

- 1 Godfrey, J. W. (1969), *The mechanism of a road network*. Traffic Engineering and Control,
2 11, 323–327.
- 3 Lamotte, R. & Geroliminis, N. (2016), *The morning commute in urban areas: Insights*
4 *from theory and simulation*. In *Transportation Research Board 95th Annual Meeting.*,
5 16–2003.
- 6 Mariotte, G., Leclercq, L. & Laval, J. A. (2017), *Macroscopic urban dynamics: Analytical*
7 *and numerical comparisons of existing models*. Transportation Research Part B, 101,
8 245–267, [doi:10.1016/j.trb.2017.04.002](https://doi.org/10.1016/j.trb.2017.04.002).
- 9 Sbayti, H., Lu, C.-C. & Mahmassani, H. S. (2007), *Efficient implementation of method*
10 *of successive averages in simulation-based dynamic traffic assignment models for large-*
11 *scale network applications*. Transportation Research Record: Journal of the Transporta-
12 tion Research Board, 2029, 22–30, [doi:10.3141/2029-03](https://doi.org/10.3141/2029-03).
- 13 Sheffi, Y. (1985), *Urban Transportation networks: Equilibrium Analysis with Mathematical*
14 *Programming Methods*, chap. 10 and 11. Prentice Hall Inc., United States of America.
- 15 Yildirimoglu, M. & Geroliminis, N. (2014), *Approximating dynamic equilibrium conditions*
16 *with macroscopic fundamental diagrams*. Transportation Research Part B: Methodolog-
17 ical, 70, 186–200, [doi:10.1016/j.trb.2014.09.002](https://doi.org/10.1016/j.trb.2014.09.002).
- 18 Yildirimoglu, M., Ramezani, M. & Geroliminis, N. (2015), *Equilibrium analy-*
19 *sis and route guidance in large-scale networks with mfd dynamics*. Transporta-
20 tion Research Part C: Emerging Technologies, 59, 404 – 420, ISSN 0968-090X,
21 [doi:https://doi.org/10.1016/j.trc.2015.05.009](https://doi.org/10.1016/j.trc.2015.05.009), special Issue on International Sympo-
22 sium on Transportation and Traffic Theory.